JavaScript is disabled on your browser.

compact1, compact2, compact3

java.io

Class FilterOutputStream

java.lang.Object
    java.io.OutputStream
        java.io.FilterOutputStream

All Implemented Interfaces:

`Closeable`, `Flushable`, `AutoCloseable`

Direct Known Subclasses:

`BufferedOutputStream`, `CheckedOutputStream`, `CipherOutputStream`, `DataOutputStream`, `DeflaterOutputStream`, `DigestOutputStream`, `InflaterOutputStream`, `PrintStream`

```
public class FilterOutputStream
extends OutputStream
```

This class is the superclass of all classes that filter output streams. These streams sit on top of an already existing output stream (the *underlying* output stream) which it uses as its basic sink of data, but possibly transforming the data along the way or providing additional functionality.

The class `FilterOutputStream` itself simply overrides all methods of `OutputStream` with versions that pass all requests to the underlying output stream. Subclasses of `FilterOutputStream` may further override some of these methods as well as provide additional methods and fields.

Since:

`JDK1.0`

Field Summary

Fields

Modifier and Type	Field	Description
protected OutputStream	out	The underlying output stream to be filtered.

Constructor Summary

Constructors

Constructor	Description
FilterOutputStream(OutputStream out)	Creates an output stream filter built on top of the specified underlying output stream.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
void	close()	Closes this output stream and releases any system resources associated with the stream.
void	flush()	Flushes this output stream and forces any buffered output bytes to be written out to the stream.
void	write(byte[] b)	Writes `b.length` bytes to this output stream.
void	write(byte[] b, int off, int len)	Writes `len` bytes from the specified `byte` array starting at offset `off` to this output stream.
void	write(int b)	Writes the specified `byte` to this output stream.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Detail

out

```
protected OutputStream out
```

The underlying output stream to be filtered.

Constructor Detail

FilterOutputStream

```
public FilterOutputStream(OutputStream out)
```

Creates an output stream filter built on top of the specified underlying output stream.

Parameters:

`out - the underlying output stream to be assigned to the field this.out for later use, or null if this instance is to be created without an underlying stream.`

Method Detail

write

```
public void write(int b)
           throws IOException
```

Writes the specified `byte` to this output stream.

The `write` method of `FilterOutputStream` calls the `write` method of its underlying output stream, that is, it performs `out.write(b)`.

Implements the abstract `write` method of `OutputStream`.

Specified by:

`write` in class `OutputStream`

Parameters:

`b - the byte.`

Throws:

`IOException - if an I/O error occurs.`

write

```
public void write(byte[] b)
           throws IOException
```

Writes `b.length` bytes to this output stream.

The `write` method of `FilterOutputStream` calls its `write` method of three arguments with the arguments `b`, `0`, and `b.length`.

Note that this method does not call the one-argument `write` method of its underlying stream with the single argument `b`.

Overrides:

`write` in class `OutputStream`

Parameters:

`b - the data to be written.`

Throws:

`IOException - if an I/O error occurs.`

See Also:

`write(byte[], int, int)`

write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes `len` bytes from the specified `byte` array starting at offset `off` to this output stream.

The `write` method of `FilterOutputStream` calls the `write` method of one argument on each `byte` to output.

Note that this method does not call the `write` method of its underlying input stream with the same arguments. Subclasses of `FilterOutputStream` should provide a more efficient implementation of this method.

Overrides:

`write` in class `OutputStream`

Parameters:

`b - the data.`

`off - the start offset in the data.`

`len - the number of bytes to write.`

Throws:

`IOException` - if an I/O error occurs.

See Also:

`write(int)`

flush

```
public void flush()
           throws IOException
```

Flushes this output stream and forces any buffered output bytes to be written out to the stream.

The `flush` method of `FilterOutputStream` calls the `flush` method of its underlying output stream.

Specified by:

`flush` in interface `Flushable`

Overrides:

`flush` in class `OutputStream`

Throws:

`IOException` - if an I/O error occurs.

See Also:

`out`

close

```
public void close()
           throws IOException
```

Closes this output stream and releases any system resources associated with the stream.

The `close` method of `FilterOutputStream` calls its `flush` method, and then calls the `close` method of its underlying output stream.

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in interface `AutoCloseable`

Overrides:

`close` in class `OutputStream`

Throws:

`IOException` - if an I/O error occurs.

See Also:

`flush()`, `out`

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms.
Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.